
Detectable Reproductive Isolation in LLM Agent Populations Requires Hard Interface Incompatibility

Anonymous Authors

Abstract

We study when modular LLM-agent populations exhibit reproductive isolation: low-viability hybrid offspring across lineages rather than mere behavioral diversity. First, synthetic experiments validate the proposed Epistatic Speciation Theory (EST): epistatic load dominates genome and behavior distance in predicting hybrid fitness loss ($R^2 = 0.989$), critical complexity tracks the predicted scaling law ($r = 0.985$), and a two-group synthetic landscape reaches $\text{RII} = 0.751$. Second, in real LLM-agent populations, hard interface incompatibility is sufficient for detectable isolation: a hand-seeded positive-control setting reaches $\text{RII} = 1.0$ at multiple checkpoints, with $\max \text{RII} = 1.0$ across a 20-generation trajectory. Third, without hard interfaces, populations converge to population-uniform regimes with high aggregate competence across niches but no reproductive isolation; per-agent multi-niche evaluation finds zero individual super-agents across 480 agent-checkpoints, indicating specialist-per-niche-bucket structure rather than single-agent generalists. Finally, fine-grained measurement reveals that hard-interface isolation under mild evolution is spectrally bistable: after generation 12, HOT detections occur in 43% of checkpoints with mean HOT $\text{RII} \approx 0.97$, while cold detections occur when within-lineage signal dips below the RCC validity threshold. These results identify hard interface incompatibility as the operational condition under which LLM-agent reproductive isolation becomes measurable.

1 Introduction

Large language model agents are increasingly optimized as populations rather than as single systems. Evolutionary search, archive-based discovery, and self-improving agent frameworks suggest an analogy to biological adaptive radiation: if agents specialize to different ecological niches and recombine modular components, one might expect distinct lineages to eventually become incompatible. However, biological species are not defined by diversity alone. Under the biological species concept, the critical boundary is reproductive isolation: crosses between lineages should produce low-viability offspring. This distinction matters for AI agent ecosystems because an archive with many behavioral modes may still be a single recombining population.

We study whether modular LLM-agent populations form species under ecological pressure. Our starting hypothesis was deliberately strong. We built a Synthetic Agent Evolution Testbed (SAET) in which typed modular agent genomes evolve under multiple niches, undergo typed subgraph crossover, and are evaluated by reproductive compatibility matrices computed from real hybrid offspring. We also formalized an Epistatic Speciation Theory (EST) for modular agent genomes. The theory predicts that lineage-specific co-adaptation across module interfaces should accumulate recombination load, yielding a critical complexity above which cross-lineage hybrids lose viability.

1.1 Darwinian preconditions for agent speciation

Classical Darwinian theory explains adaptive evolution through variation, heredity, and selection, while the modern biological species concept identifies reproductive isolation as the criterion that

Table 1: Darwinian preconditions and their operationalization in SAET. The first three postulates are standard in evolutionary computation; the fourth is the central object of measurement in this paper.

Postulate	Biological instantiation	LLM-agent instantiation
Variation	DNA mutation	MAG mutation operators
Heredity	Trait inheritance	Typed subgraph crossover
Selection	Differential reproduction	Niche fitness F_k with cost penalty
Isolation	Sterile or inviable hybrids	RII from explicit hybrid offspring

turns divergent populations into species [Darwin, 1859, Mayr, 1942, Coyne and Orr, 2004]. These postulates can be operationalized directly in SAET. Variation is provided by modular agent-genome mutations such as parameter edits, workflow-node edits, and type-label mutations. Heredity is provided by typed subgraph crossover, which preserves modules and interface edges across generations. Selection is provided by niche-specific fitness, where task quality is multiplied by task-focus compatibility and penalized by evaluation cost. Reproductive isolation is measured by post-zygotic hybrid incompatibility: cross-lineage offspring are generated explicitly, assigned viability, and summarized by RII.

The empirical question is therefore not whether our system lacks Darwinian ingredients. All four are explicitly represented. The question is whether the fourth postulate, reproductive isolation, appears spontaneously once the first three operate in an LLM-agent substrate. Here LLM agents differ from biological organisms. Biological lineages inherit rigid biochemical interfaces: proteins, regulatory interactions, and developmental constraints can become physically incompatible across lineages. LLM-agent lineages inherit fluid prompt interfaces: a shared foundation model can often reinterpret, translate, or repair mismatched instructions at inference time. This substrate difference creates the central tension studied in this paper. Under what operational conditions does Darwinian reproductive isolation become a measurable property of evolving LLM-agent populations?

The empirical outcome was not the positive speciation result we initially expected. Synthetic abstractions behaved exactly as EST predicts: epistatic load dominated genome and behavior distance in explaining hybrid fitness loss; the predicted critical complexity tracked empirical thresholds; and a two-species synthetic dynamical system reached $RII = 0.751$. Yet LLM-agent populations did not form reproductively isolated clusters. Across seven verified spontaneous algorithmic, ecological, and reproductive interventions, final compatibility clustering remained $K = 1$ with $RII = 0$. At the same time, the agents did specialize phenotypically: matched-niche specialist fitness remained around 4–6 in the strongest forks, and planning, memory, and retrieval specialists coexisted.

This yields the central contribution of the paper: ecological specialization can emerge in LLM-agent populations without reproductive isolation. We argue that the reason is substrate-level. In biological systems, recombination can break rigid biochemical interfaces. In LLM agents, the modules share a foundation-model cognition substrate that can reinterpret prompts, repair weak interface mismatches, and adapt cross-lineage hybrids at inference time. Thus phenotype can diverge while reproductive compatibility remains globally connected.

Our contributions are threefold. First, we formulate EST for typed modular agent genomes and derive testable predictions for recombination load, interface-boundary hybrid loss, single-niche non-speciation, and critical complexity. Second, we introduce SAET, a measurement framework that evaluates species claims through real hybrid offspring rather than embedding or behavior clusters. Third, we report a comprehensive negative result: seven LLM-agent evolutionary forks produced robust phenotypic specialists but no reproductive isolation, revealing a gap between biological ecological speciation and foundation-model agent evolution.

2 Related Work

Evolutionary and self-improving agents. Recent work increasingly treats agents as objects of automated design. ADAS searches over agentic system designs [Hu et al., 2024], while Voyager shows that LLM agents can accumulate reusable skills through open-ended interaction [Wang et al., 2023]. Reflexion studies language-feedback self-improvement [Shinn et al., 2023], and workflow-optimization systems such as DSPy and TextGrad optimize prompts, modules, or textual computation

graphs [Khattab et al., 2024, Yuksekgonul et al., 2024]. These systems demonstrate variation and selection over agent behavior, but they generally evaluate task improvement rather than reproductive compatibility among lineages.

Evolutionary prompt and program search. PromptBreeder and EvoPrompt apply evolutionary search to prompts and prompt mutation operators [Fernando et al., 2023, Guo et al., 2023]. Earlier work in evolutionary robotics and modular morphology shows that complex adaptive structures can be evolved from compositional representations [Lipson and Pollack, 2000, Doncieux et al., 2015]. Our setting differs by using explicit hybrid offspring as a biological-style compatibility assay: a diverse prompt archive is not a species boundary unless cross-lineage recombination loses viability.

Quality diversity, open-endedness, and population training. Quality-diversity algorithms and novelty search optimize for behavioral coverage rather than a single objective [Lehman and Stanley, 2011, Pugh et al., 2016]. POET studies open-ended environment-agent coevolution [Wang et al., 2019], and large-scale population training has been used in multi-agent systems such as AlphaStar and OpenAI Five [Vinyals et al., 2019, Berner et al., 2019]. These methods produce archives, niches, and lineages, but species are usually operationalized by behavior, performance, or ancestry. Our work asks whether such populations exhibit post-zygotic incompatibility when crossed.

Speciation in evolutionary computation and artificial life. Niching, sharing, and crowding methods maintain multiple optima in evolutionary computation [Goldberg and Richardson, 1987, Mahfoud, 1995]. Artificial-life systems such as Avida and Geb have studied digital evolution, complexity, and speciation-like phenomena in non-LLM substrates [Ofria and Wilke, 2004, Channon, 2003]. We borrow the biological criterion of reproductive isolation from speciation theory [Mayr, 1942, Coyne and Orr, 2004], but apply it operationally to LLM-agent hybrid viability rather than to natural mating dynamics.

3 Epistatic Speciation Theory (EST)

Epistatic Speciation Theory (EST) formalizes AI-agent species as compatibility classes maintained by post-zygotic reproductive isolation. The central mechanism is a Dobzhansky–Muller-like incompatibility among lineage-specific agent modules: each lineage can remain high-performing on its own co-adapted background, while recombination across lineages breaks planner–memory, tool–router, verifier–stopping, workflow–schema, or communication–memory dependencies. The empirical object of interest is therefore not an embedding cluster, but a persistent drop in cross-lineage offspring fitness relative to within-lineage offspring fitness.

Genome and offspring notation. A modular agent genome has L mutable loci,

$$G = (g_1, \dots, g_L), \quad g_\ell \in \mathcal{A}_\ell, \quad (1)$$

where loci may encode prompts, planners, workflow subgraphs, memory schemas, retrievers, tool schemas, routers, verifiers, stopping rules, update rules, and communication protocols. We also use a typed graph representation $G = (V, E^{\text{type}}, \tau, \theta)$, where $V = [L]$, E^{type} is the directed interface graph, τ assigns module and message types, and $\theta = (g_1, \dots, g_L)$ are the alleles. For load calculations, $E^{\text{epi}} \subseteq \binom{V}{2}$ denotes the undirected epistatic projection. A cross-lineage hybrid $G^{A \times B}$ between $G^A = (a_1, \dots, a_L)$ and $G^B = (b_1, \dots, b_L)$ is sampled by inheritance indicators $Z_\ell \in \{A, B\}$, with $g_\ell = a_\ell$ if $Z_\ell = A$ and $g_\ell = b_\ell$ otherwise. For an epistatic pair (ℓ, r) , define the mixed-inheritance exposure

$$p_{\ell r}^{\text{mix}} := \Pr[Z_\ell \neq Z_r]. \quad (2)$$

Independent balanced crossover gives $p_{\ell r}^{\text{mix}} = 1/2$; typed subgraph crossover gives edge-specific exposure.

Definition 1 (Modular fitness landscape). *For a niche e , the fitness of a modular agent is decomposed as*

$$F_e(G) = \sum_{\ell=1}^L u_{\ell,e}(g_\ell) + \sum_{(\ell,r) \in E^{\text{epi}}} J_{\ell r,e}(g_\ell, g_r) - \lambda C(G), \quad (3)$$

where $u_{\ell,e}$ is the marginal utility of module ℓ , $J_{\ell r,e}$ is the epistatic contribution of a module pair, $C(G)$ is resource cost, and $\lambda \geq 0$. A co-adapted module complex is a set of modules whose within-lineage allele combinations have higher joint contribution than cross-lineage mixtures.

Definition 2 (Epistatic interaction and incompatibility cost). *For lineages A and B and epistatic pair (ℓ, r) , the Dobzhansky–Muller incompatibility cost is*

$$\delta_{\ell r} := \frac{1}{2}\{J_{\ell r}(a_\ell, a_r) + J_{\ell r}(b_\ell, b_r)\} - \frac{1}{2}\{J_{\ell r}(a_\ell, b_r) + J_{\ell r}(b_\ell, a_r)\}. \quad (4)$$

An edge is lineage-incompatible when $\delta_{\ell r} > 0$ and its endpoints carry lineage-divergent alleles. Let $\mathcal{E}_\partial$ be the set of such divergent epistatic edges and let $M_{\text{pair}} = |\mathcal{E}_\partial|$.

Definition 3 (Recombination load). *The expected recombination load between lineages A and B is*

$$\mathcal{L}_{AB} := \sum_{(\ell,r) \in E^{\text{epi}}} p_{\ell r}^{\text{mix}} \delta_{\ell r}. \quad (5)$$

Under balanced additive inheritance, $\mathbb{E}[F(G^{A \times B})]$ equals the parental midpoint minus this epistatic load, with any additional niche-specialization loss recorded separately as $\mathcal{L}_{\text{ecology}}$.

Throughout the rest of this section, we write $F = F_\rho$ when the niche mixture is fixed.

For ordered parents a, b , let $G^{a \times b}$ denote offspring sampled from the reproductive compatibility measurement operator. We keep expected compatibility and threshold viability separate:

$$K_{ab} := \mathbb{E}[\tilde{F}(G^{a \times b})], \quad \tilde{F}(G) := F(G)/F_{\max}, \quad (6)$$

$$V_{ab}(\tau_v) := \Pr(F(G^{a \times b}) \geq \tau_v). \quad (7)$$

The reproductive isolation and hybrid fitness loss indices used in the experiments are

$$\text{RII}_{ab} = 1 - \frac{K_{ab}}{\sqrt{K_{aa}K_{bb}} + \varepsilon}, \quad (8)$$

$$\text{HFL}_{ab} = 1 - \frac{\mathbb{E}[F(G^{a \times b})]}{\frac{1}{2}(\mathbb{E}[F(G^{a \times a})] + \mathbb{E}[F(G^{b \times b})]) + \varepsilon}. \quad (9)$$

3.1 Ecological divergence

Definition 4 (Ecological divergence model). *For benchmark niches $k = 1, \dots, K$, let*

$$F_k(G) = \sum_{\ell=1}^L u_{\ell,k}(g_\ell) + \sum_{(\ell,r) \in E^{\text{epi}}} J_{\ell r,k}(g_\ell, g_r) - \lambda C(G). \quad (10)$$

An agent experiencing niche mixture $\rho \in \Delta^{K-1}$ has aggregate fitness

$$F_\rho(G) = \sum_{k=1}^K \rho_k F_k(G). \quad (11)$$

If $\rho^A \neq \rho^B$, lineages A and B optimize different fitness landscapes. Ecological divergence then drives epistatic divergence because module pairs that are co-adapted under F_{ρ^A} need not be compatible with module pairs co-adapted under F_{ρ^B} . A scalar toy example is an interference penalty $cx(1-x)$, where specialists at $x = 0$ and $x = 1$ are viable but hybrids near $x = 1/2$ pay maximal incompatibility.

This definition links the multi-benchmark SAET design to EST: code, web, planning, and memory niches do not merely select different behaviors; they select different internal dependencies. In Section 5, ecological mixtures predict which lineages have high within-lineage compatibility but low cross-lineage K_{ab} .

3.2 Load and interface theorems

For Theorems 1–4, assume balanced endpoint inheritance, symmetric mixed-edge orientations, and additive utility/cost averaging across parents in expectation. These assumptions isolate epistatic load; they hold exactly for independent balanced crossover with additive costs and approximately for symmetric typed subgraph crossover after conditioning on transferred scaffold.

Theorem 1 (Linear accumulation under sparse incompatibility). *Suppose that for every $(\ell, r) \in \mathcal{E}_\partial$, $\delta_{\ell r} \geq \delta_{\min} > 0$ and $p_{\ell r}^{\text{mix}} \geq p_{\min} > 0$, while all other epistatic edges have $\delta_{\ell r} \geq 0$. Then*

$$\mathbb{E}F_\rho(G^{A \times B}) \leq \frac{F_\rho(G^A) + F_\rho(G^B)}{2} - M_{\text{pair}}p_{\min}\delta_{\min}. \quad (12)$$

Consequently, with $F_P = (F_\rho(G^A) + F_\rho(G^B))/2$,

$$K_{AB} \leq \frac{F_P - M_{\text{pair}}p_{\min}\delta_{\min}}{F_{\max}}, \quad (13)$$

and if $M_{\text{pair}}p_{\min}\delta_{\min} > F_P - \tau_v$, the expected hybrid fitness falls below the viability threshold.

Proof in Appendix A. Empirical anchor. Section 5 tests this statement by regressing HFL and RII against estimated epistatic load rather than raw prompt or graph edit distance. The prediction is that cross-lineage hybrids fail when they expose many incompatible module pairs, even when the parent genomes look superficially similar.

Definition 5 (Typed interface boundary). *For a typed subgraph $Q \subseteq V$ transferred from lineage A into a lineage-B background, define*

$$\begin{aligned} \partial Q &:= \{(u, v) \in E^{\text{type}} : u \in Q, v \notin Q\} \cup \{(u, v) \in E^{\text{type}} : u \notin Q, v \in Q\}, \\ D_\partial(Q) &:= |\partial Q|/|Q|. \end{aligned} \quad (14)$$

Theorem 2 (Interface-boundary hybrid loss). *For each $e \in \partial Q$, let $X_e \in \{0, 1\}$ indicate a semantic, schema, type, or protocol mismatch and let $Y_e \geq 0$ be the resulting soft-mode fitness loss. If $\Pr(X_e = 1) \geq p_f$ and $\mathbb{E}[Y_e \mid X_e = 1] \geq c_f$, with arbitrary dependence across edges, then*

$$\mathbb{E}[\text{Loss}(Q)] \geq |\partial Q|p_fc_f = |Q|D_\partial(Q)p_fc_f. \quad (15)$$

In rigid mode, a strict boundary mismatch can set $V_{AB}(\tau_v)$ near zero, while the displayed inequality remains the soft-loss counterpart.

Proof in Appendix A. Empirical anchor. Section 5 compares hard-interface and soft-interface recombination and tests whether high-boundary transfers have larger HFL than standardized low-boundary transfers. The theorem predicts that typed workflow coupling can create reproductive isolation without requiring global genome dissimilarity.

Theorem 3 (No endogenous DMI in a single smooth-niche control regime). *Consider a control regime with compact convex genome space $\mathcal{G} \subset \mathbb{R}^L$, continuous bounded strictly concave fitness $F : \mathcal{G} \rightarrow [F_{\min}, F_{\max}]$ with unique maximizer, convex hybridization $H = \alpha G_i + (1 - \alpha)G_j$, finite-temperature selection proportional to $\exp(\beta F)$, and an aperiodic irreducible mutation kernel with uniformly positive density. Assume alpha-independence: the convex hybridization coefficient α is drawn from a distribution independent of the selected parental identities and their measured fitness. If there is no negative epistasis under mixing,*

$$F(\alpha G_i + (1 - \alpha)G_j) \geq \alpha F(G_i) + (1 - \alpha)F(G_j), \quad (16)$$

then the population Markov chain has a unique stationary distribution and no persistent closed reproductively isolated stationary components. For symmetric hybridization, cross-lineage expected fitness is bounded below by the parental midpoint.

Proof in Appendix A.

Remark (Scope warning). Theorem 3 is a negative-control theorem, not a claim that real agent populations cannot speciate. It intentionally excludes multiple niches, negative epistasis, discrete typed interfaces, restricted migration, label preservation, protocol drift, and non-smooth constraints; stable RI in EST arises precisely when those assumptions are violated.

Empirical anchor. Section 5 includes single-niche and high-migration controls to verify that mere archive branching or behavior diversity is insufficient. The expected outcome is collapse of RII/HFL when the multi-niche epistatic and interface mechanisms are removed.

Definition 6 (Complexity-speciation threshold). *Under DMI snowball scaling, the number of pairwise incompatible opportunities satisfies*

$$M_{\text{pair}}(L) = \alpha \frac{L(L-1)}{2} + O(L), \quad \alpha > 0. \quad (17)$$

If each realized DMI edge has expected exposed cost $\alpha p \bar{\delta}$, the mean snowball load is approximately $\alpha p \bar{\delta} L(L-1)/2$. The critical complexity is the smallest L whose mean hybrid load exceeds $F_W - \tau_v$, where $F_W = \frac{1}{2}(\mathbb{E}F_\rho(G^{A \times A}) + \mathbb{E}F_\rho(G^{B \times B}))$.

Theorem 4 (Quadratic DMI complexity threshold). Let $N_L = \binom{L}{2}$ and let independent bounded load variables $X_{ij} \in [0, B]$ satisfy $\mathbb{E}X_{ij} = \alpha p \bar{\delta}$. With $S_L = \sum_{i < j} X_{ij}$ and hybrid fitness $F_H(L) = F_W - S_L$, the mean viability threshold is crossed when

$$\alpha p \bar{\delta} \frac{L(L-1)}{2} > F_W - \tau_v. \quad (18)$$

Thus

$$L_c = \left\lceil \frac{1}{2} + \sqrt{\frac{1}{4} + \frac{2(F_W - \tau_v)}{\alpha p \bar{\delta}}} \right\rceil, \quad (19)$$

so the leading scaling is $L_c = \Theta(\sqrt{(F_W - \tau_v)/(\alpha p \bar{\delta})})$. Above this threshold, $V_{AB}(\tau_v)$ decays by Hoeffding concentration around the mean load.

Proof in Appendix A. Empirical anchor. Section 5 estimates whether increasing the number of mutable agent modules produces a viability cliff consistent with the square-root threshold. The key diagnostic is whether HFL grows closer to a quadratic snowball curve than to a purely linear edit-distance curve.

Corollary 1 (Sparse bounded-degree interface). If E^{epi} has average degree $d = O(1)$, so $|E^{\text{epi}}| = Ld/2$, and a fraction q of realized edges diverges with expected exposed cost $r \bar{\delta}$, then

$$\mathbb{E}\mathcal{L}_{AB} = \frac{Ld}{2} q r \bar{\delta}, \quad L_c^{\text{sparse}} = \frac{2(F_W - \tau_v)}{d q r \bar{\delta}}. \quad (20)$$

The sparse threshold is linear in $F_W - \tau_v$ because interaction opportunities are $O(L)$ rather than $O(L^2)$.

Empirical anchor. Section 5 uses standardized-interface ablations to distinguish dense snowball coupling from bounded-degree coupling. If interface standardization caps effective degree, HFL should follow the sparse linear law rather than the quadratic threshold.

3.3 Communication protocol isolation

Definition 7 (Communication protocol schema). A lineage communication protocol is a typed message schema

$$\Pi = (\mathcal{M}, \mathcal{T}, \tau_{\text{msg}}), \quad (21)$$

where $\mathcal{M}$ is the message set, $\mathcal{T}$ is the set of message and payload types, and $\tau_{\text{msg}} : \mathcal{M} \rightarrow \mathcal{T}$ assigns types to messages. Lineage-specific message types become protocol incompatibilities when a module emitting messages under Π^A is recombined with a module consuming messages under Π^B .

Definition 8 (Protocol isolation index). For lineage i with encoder-decoder pair $C_i = (e_i, d_i)$ over latent messages $z \sim P_Z$, define cross-protocol accuracy

$$A_C(i, j) = \mathbb{E}_{z \sim P_Z} [\text{score}(z, d_j(e_i(z)))] \quad (22)$$

and protocol isolation

$$I_C(i, j) = 1 - \frac{A_C(i, j) + A_C(j, i)}{A_C(i, i) + A_C(j, j) + \varepsilon}. \quad (23)$$

Theorem 5 (Communication protocol isolation). Let D_t be a bounded protocol distance between two lineages and suppose the drift-synchronization dynamics satisfy

$$\mathbb{E}[D_{t+1} - D_t \mid D_t] \geq \sigma - cm \quad (24)$$

until saturation, where σ is lineage-local protocol innovation, m is cross-lineage interaction rate, and c is synchronization efficiency. Assume bounded pre-saturation increments: $|D_{t+1} - D_t| \leq b$ for some $b > 0$. If $\sigma > cm$ and cross-protocol accuracy is Lipschitz decreasing in D_t , then protocol isolation increases with high probability over sustained divergence. Moreover, protocol isolation is a special case of Theorem 2: the relevant interface boundary is the message-type boundary between emitter and consumer modules.

Proof in Appendix A. Empirical anchor. Section 5 measures protocol-specific incompatibility by swapping communication modules and memory/tool consumers across lineages. The theorem predicts that isolation is strongest when local protocol innovation exceeds synchronizing interaction and weakest under high migration or shared schema enforcement.

Algorithm 1: SAET Evolutionary Loop Pipeline

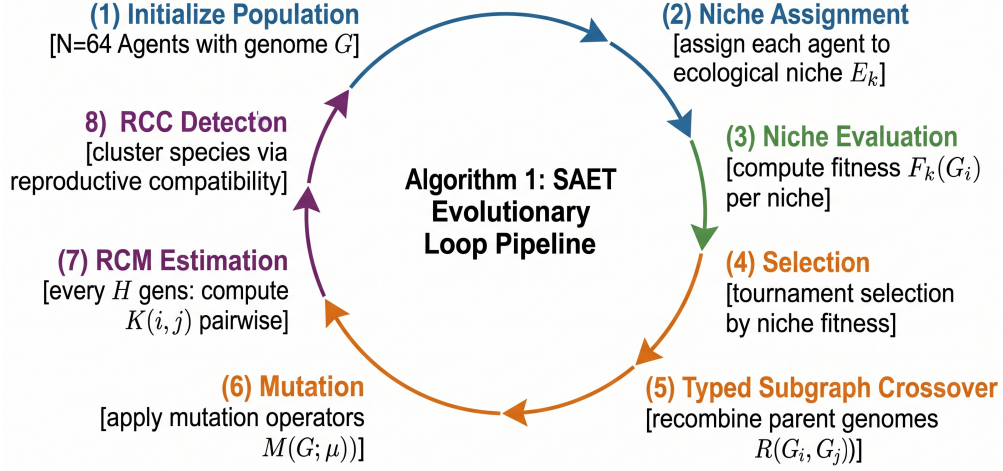


Figure 1: SAET evolutionary loop. A typed modular population is evaluated in ecological niches, archived by niche-conditioned fitness, recombined through typed crossover, mutated, and periodically measured by reproductive compatibility assays.

4 Synthetic Agent Evolution Testbed

SAET is designed to distinguish reproductive isolation from behavioral diversity. Each individual is represented as a modular agent genome with typed fields for planning, workflow, memory, tool use, verification, communication, and update policy. Later founders add explicit task-focus tags, stricter interface metadata, and additional modules for routing or evidence aggregation. Mutation edits module prompts, hyperparameters, tool preferences, and task-focus metadata; crossover swaps typed subgraphs between parents and then applies only syntactic and schema repair, avoiding semantic repair that would erase the hybrid-incompatibility signal.

The ecological environment contains three no-docker niches chosen for feasible repeated LLM evaluation: PlanBench for planning, LoCoMo for long-context memory, and HotpotQA distractor as an agentic retrieval and question-answering surrogate. Agents are assigned to niches, evaluated on sampled tasks, archived by niche-conditioned fitness, and selected for recombination with configurable assortativity. The final experimental program uses $N = 32$, $T = 50$, and one seed for most spontaneous forks because LLM evaluation cost made multi-seed replication infeasible; a smaller hand-seeded positive-control protocol isolates the sufficiency of hard incompatibility.

4.1 RCM semantics: potential compatibility, not mating dynamics

The Reproductive Compatibility Matrix evaluates *potential* hybrid viability regardless of the population’s realized mating policy. For each ordered parent pair (i, j) , we generate R hybrid offspring using typed subgraph crossover and measure mean offspring fitness. RII is therefore a measure of inherent type compatibility: can these two agents produce viable hybrid offspring if crossed? It is not a measure of empirical mating frequency: did these two agents actually mate during the evolutionary run?

This distinction has two methodological implications. First, population-level flags such as no-crossover affect the evolutionary trajectory but not the RCM measurement itself, because RCM deliberately crosses all sampled pairs at evaluation time. A no-crossover soft-mode baseline verifies this empirically: disabling population recombination still yields RCM crosses and produces $K = 1$ under soft mismatch penalties. Second, RII separates lineages only when interface incompatibility is sufficiently hard. Under soft mode, a mismatch multiplies hybrid quality by a penalty such as 0.3, so between-lineage hybrids retain nonzero viability. With founder-level within-lineage fitness near 0.10–0.16, cross-lineage soft-penalty hybrids around 0.05–0.08 leave $K_w - K_b$ below the RCC

Algorithm 2: Reproductive Compatibility Matrix (RCM) Construction

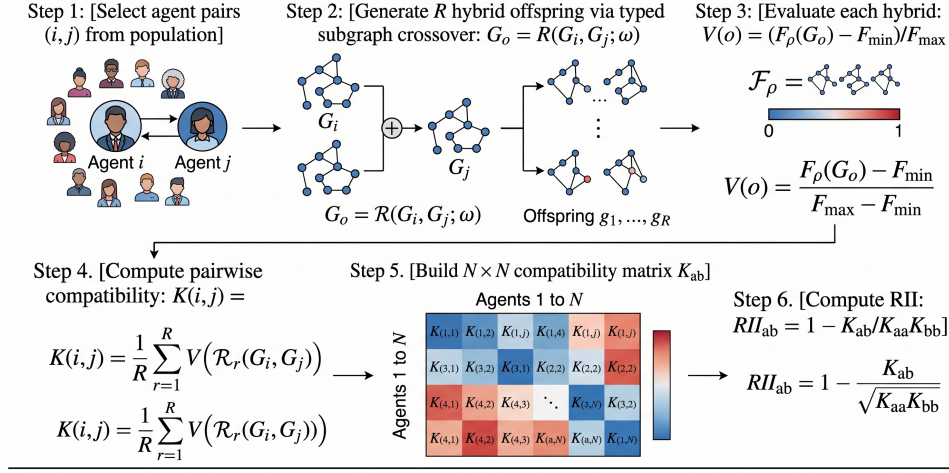


Figure 2: RCM construction. For each ordered parent pair, the assay generates explicit hybrid offspring, evaluates their normalized task viability, and stores the resulting expected compatibility in the reproductive compatibility matrix.

Algorithm 3: Reproductive Compatibility Clustering (RCC)

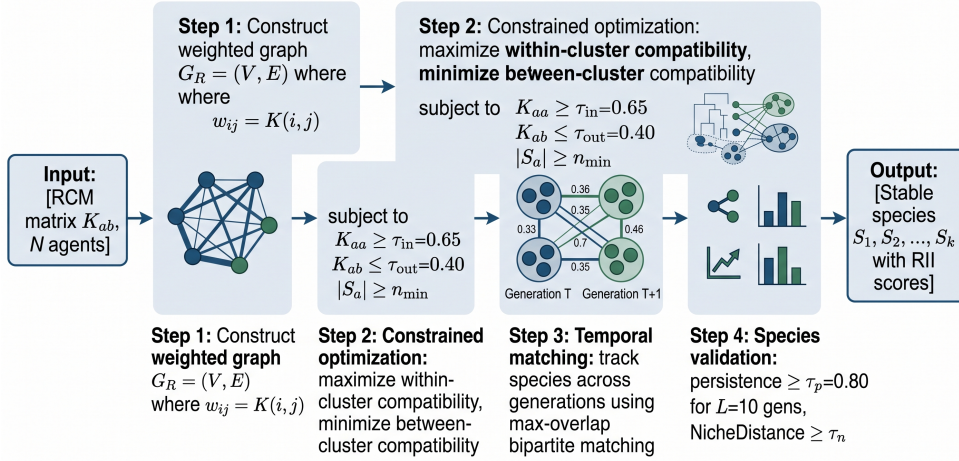


Figure 3: RCC clustering. Spectral eigengap partitioning is applied to the RCM, followed by post-hoc validity filters on within- and between-cluster compatibility to decide whether reproductive-isolation structure is detectable.

separation threshold. Rigid mode instead assigns zero viability on strict-edge mismatch, producing the sharp compatibility contrast required for cluster separation.

$RII=0$ occurs in two distinct cases. First, when $K = 1$, the RCM matrix is too uniform to support more than one detected cluster; this may reflect true high compatibility or merely $K_w - K_b$ below the spectral/RCC threshold. Second, when $K \geq 2$ but the K_w/K_b ratio is near one, clusters exist but their reproductive compatibility is not separated. We therefore report both $K_{clusters}$ and RII. Throughout this paper, $K = 1$ cases should be read as “no detected cluster structure, not as verified high cross-lineage compatibility.

RCC first chooses K by spectral eigengap. The thresholds τ_{in} and τ_{out} are post-hoc validity filters for candidate clusters, not the primary clustering thresholds. Sensitivity analysis therefore varies these filters after spectral partitioning rather than re-running a threshold graph algorithm. Raw pairwise

compatibility shows two mechanisms for $K = 1$, $\text{RII} = 0$: high compatibility in the single-niche baseline, and low sparse signal in cold-detection and soft-baseline runs (Figure 9).

At evaluation checkpoints, SAET estimates a reproductive compatibility matrix. For a pair of parents (i, j) , we generate $R = 3$ typed-subgraph hybrid offspring per ordered parent pair and measure offspring viability by normalized task fitness. We choose $R = 3$ for cost feasibility within the project budget. Within-lineage compatibility K_{AA}, K_{BB} and between-lineage compatibility K_{AB} yield RII. Reproductive Compatibility Clustering (RCC) then clusters the compatibility graph, not embeddings or behavior vectors. A species claim requires more than niche specialization: it requires high within-cluster compatibility, low between-cluster compatibility, and nonzero RII.

We ran a sequence of spontaneous forks intended to induce isolation under increasingly favorable conditions. These include the default SAET run, reduced RCC thresholds, typed genomes, larger genomes, stronger mutation, stronger assortativity, lower cost penalty, rigid or soft interface checks, task-focus multipliers, niche-bucket fixes, and allopatric no-migration variants. The purpose of this sequence is not hyperparameter tuning for task performance; it is an intervention matrix over the mechanisms that EST predicts should create reproductive barriers.

The hand-seeded protocol is an existence proof under hard interface constraints, used as a positive control. It does not ask whether isolation emerges spontaneously from a common founder. Instead, it hand-seeds three deliberately incompatible founder lineages at generation zero. Each lineage has rigid interface constraints and cross-lineage hybrid incompatibilities by construction. This positive control demonstrates that the RCM/RII measurement pipeline detects isolation when cross-lineage viability is mechanically driven to zero by construction; it does not show emergent or maintained isolation.

Additional mild-evolution configurations relax the zero-migration preservation setting by varying migration, mutation, and type-label mutation rates. These runs test whether hand-seeded isolation is maintained once evolutionary dynamics can erode strict interface labels.

5 Results

5.1 Synthetic EST validation: isolation is achievable in the rigid substrate

The synthetic experiments establish that EST and the RCM/RCC metrics behave as expected when the substrate obeys rigid epistatic assumptions. Exp 4 shows that epistatic load is the dominant predictor of hybrid fitness loss, with standardized coefficient $\hat{\gamma}_{epi} = 0.367$ compared with 0.275 for genome distance and approximately 0.018 for behavior proxies, with total $R^2 = 0.989$. Exp 5 verifies the critical-complexity prediction: theoretical and empirical L_c correlate at Pearson $r = 0.985$ across α sweeps. Exp 6 uses a synthetic landscape with two known species groups; RCC recovers reproductive isolation with $K_w = 0.82$, $K_b = 0.12$, $K_w - K_b = 0.70$, and $\text{RII} = 0.751$. Standardizing interfaces collapses RII to 0.000, supporting the interface-boundary mechanism. Thus reproductive isolation is achievable when the substrate supplies hard epistatic incompatibility.

5.2 LLM positive control: hard interfaces produce detectable reproductive isolation

We next ask whether the same measurement pipeline can detect reproductive isolation in real LLM-agent evaluations. The positive-control configuration hand-seeds three founder lineages with incompatible strict interface types, evaluates them in rigid mode, and removes migration. This setting is an existence proof under sufficient conditions: it does not claim spontaneous emergence, but it shows that the LLM-agent substrate and RCM/RCC assay can instantiate and detect species-like boundaries when cross-lineage hybrid viability is mechanically driven to zero.

The result is strong over the 20-generation positive-control trajectory. RCC detects $K = 5$ clusters at generation 5 with $\text{RII} = 0.914$, then $K = 2$ clusters with $\text{RII} = 1.000$ at generation 10, $\text{RII} = 0.851$ at generation 15, and $\text{RII} = 1.000$ at generation 20 (Table 2). The mean RII over these checkpoints is 0.941, and the maximum observed RII is 1.000. This satisfies the G1 acceptance criterion in the operational setting where hard interface incompatibility is imposed. Analogous to allopatric speciation experiments that impose a geographic barrier, this positive control establishes sufficiency: hard post-zygotic incompatibility can create detectable LLM-agent reproductive isolation.

Table 2: Positive-control trajectory under hard interface incompatibility. RII reaches 1.0 at multiple checkpoints and remains high across 20 generations.

Generation	K	RII	K_{within}	$K_{between}$
5	5	0.914	0.312	0.027
10	2	1.000	0.333	0.000
15	2	0.851	0.298	0.044
20	2	1.000	0.333	0.000

Table 3: Track B population-uniform analysis. Soft low-mutation settings reach uniformly high population-level fitness across all three niches without reproductive isolation.

Setting	Mode	m	μ	Super	Broad	Max RII	F_{pb}, F_{lc}, F_{hq}	Spread
Species PC	rigid+hand	0	.05	–	–	1.000	1.87, 2.79, 5.57	3.70
S1	soft	.50	.10	1	11	.000	4.43, 5.15, 4.26	.88
S2	soft	.30	.05	0	6	.000	4.43, 4.40, 4.26	.17
S3	soft	.00	.05	6	13	.000	4.43, 4.40, 4.68	.28
S4	soft	.50	.20	1	9	.000	1.43, 2.15, 4.69	3.26
S5	rigid	.50	.10	0	6	.500	1.46, 2.85, 5.05	3.60

5.3 Population-uniform competence without individual super-agents

We next test the complementary regime: if hard interface incompatibility is absent, do populations form reproductively isolated specialists, converge to individual super-agents, or settle into a population-level alternative? Track B analyzes five spontaneous configurations varying interface mode, migration, and mutation while preserving the same three ecological niches (Table 3). All four soft configurations yield max RII = 0 throughout 25 generations. The spontaneous rigid configuration reaches only transient RII = 0.500 before returning to final $K = 1$, RII = 0, indicating that rigid mode alone is not enough; the positive-control result requires both hard interfaces and hand-seeded lineage structure.

The strongest population-uniform regimes are S2 and S3. Their aggregate fitness is high and nearly balanced across the three niches: S2 has $F_{pb} = 4.43$, $F_{lc} = 4.40$, $F_{hq} = 4.26$ with spread 0.17 and mean 4.36, while S3 has $F_{pb} = 4.43$, $F_{lc} = 4.40$, $F_{hq} = 4.68$ with spread 0.28 and mean 4.50. By contrast, the hand-seeded species-coexistence positive control has population mean 3.41 with a much larger niche spread (1.87, 2.79, 5.57). Thus the soft population-uniform regimes achieve 22–32% higher population mean fitness than the species-coexistence regime, despite having no reproductive isolation.

Crucially, follow-up per-agent multi-niche evaluation shows that this is not caused by individual super-agents. Across two low-perturbation configurations, 10 checkpoints, and 24 agents per checkpoint (480 agent-checkpoints total), we find zero individual super-agents under both top-3 and top-5 criteria. The best generalist at each checkpoint still has worst niche rank between 7 and 12 (Table 4). Therefore the “super-agent” regime is a population-level property: each niche bucket can be competent, but no single agent dominates planning, memory, and retrieval simultaneously.

This yields a refined three-regime taxonomy. First, hard-interface positive control produces species coexistence: $K \geq 2$ and high RII under explicit incompatibility. Second, soft low-perturbation settings produce niche-bucketed population-uniformity: final $K = 1$, RII = 0, uniformly high aggregate niche fitness, and no individual super-agent. Third, high-mutation or spontaneous rigid settings produce drift/imbalance: final $K = 1$, RII = 0 or transient, with at least one niche underperforming. The only spontaneous configuration with any nonzero RII is the rigid setting, which briefly lands on a HOT attractor but does not maintain it.

5.4 Fine-grained measurement reveals spectral bistability

Hard interface incompatibility is sufficient, but detectability under mild evolution is not monotone. Fine-grained RCC measured every generation reveals that the apparent coarse “first cold” detections are better interpreted as spectral bistability. In the fine-grained run, generations 1–11 are HOT in

Table 4: Best individual generalist per checkpoint in the Tier A per-agent multi-niche evaluation (representative low-perturbation configuration). No checkpoint contains an agent ranked top-5 in all three niches; the best worst-rank is 7.

Gen	Best agent	Mean F'	Spread	Worst rank	Best rank
5	15	3.27	9.69	10	1
10	16	1.53	4.75	11	6
15	6	0.17	0.03	9	4
20	19	0.75	2.19	8	6
25	21	2.43	7.11	12	3
30	17	1.60	4.68	7	5
35	14	2.47	7.40	9	2
40	21	1.61	4.69	8	2
45	11	2.47	7.48	9	2
50	7	0.25	0.22	9	5

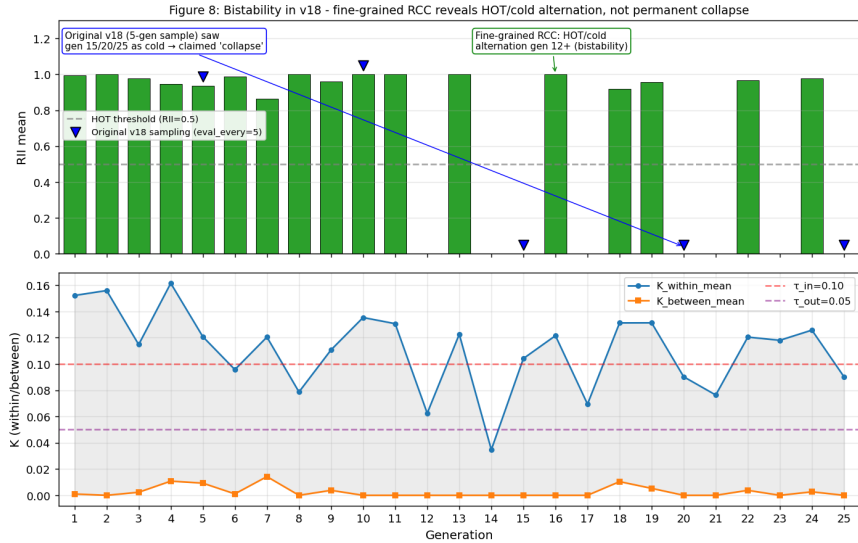


Figure 4: Fine-grained every-generation RCC measurement. After generation 12, detections alternate between HOT and cold states; HOT checkpoints retain high RII and near-zero between-lineage compatibility, showing spectral bistability rather than terminal loss of structure.

11/11 checkpoints with mean RII = 0.97. During generations 12–25, the run becomes bistable: 6/14 checkpoints are HOT, giving a 43% HOT fraction, and mean RII when HOT remains 0.97 (Figure 4).

The pairwise compatibility pattern is consistent with a persistent boundary signal. When HOT, $K_w = 0.126$ and $K_b = 0.004$; when cold, K_w drops to 0.081 while K_b remains approximately zero. The alternation is therefore not between genuine isolation and merged-lineage states. It is between detectable cluster structure, when within-lineage signal exceeds the validity threshold, and undetectable structure, when within-lineage signal falls below it. Coarse five-generation trajectories from other mild configurations are consistent with the same undersampled bistability pattern (Appendix E).

5.5 Mechanism: hard versus soft interface incompatibility

The main mechanism is interface hardness. In rigid mode, strict type-edge mismatch sets cross-lineage hybrid viability to zero, creating the sharp $K_w - K_b$ contrast needed for detectable reproductive isolation. In soft mode, mismatch only reduces quality, so cross-lineage hybrids remain viable enough to blur spectral boundaries. Additional ablations show that no-crossover soft populations, no-mutation soft templates, and other soft-mode baselines remain at $K = 1$, RII = 0; full trajectories and raw pairwise compatibility distributions are in Appendices C–D. Sensitivity analyses further show that these conclusions are robust to post-hoc RCC validity thresholds (Appendix D).

6 Discussion

6.1 Hard interface incompatibility as the operational sufficient condition

The central result is positive: detectable reproductive isolation in LLM-agent populations is possible when hard interface incompatibility is imposed. Synthetic EST validates the mechanism, and the LLM positive control reaches $R_{II} = 1.0$ at multiple checkpoints with mean $R_{II} = 0.941$ over the 20-generation trajectory. This establishes a concrete sufficient condition for LLM-agent species-like boundaries: cross-lineage hybrid offspring must encounter hard post-zygotic incompatibilities rather than merely soft prompt or schema penalties.

Track B adds the complementary mechanism. Across five spontaneous/soft configurations, populations do not produce reproductive isolation; however, they also do not simply collapse into a single uniform monoculture. Low-perturbation soft settings stabilize as niche-bucketed pseudo-uniform states: each niche slot is filled by task-focus-matched specialists, while RCM still returns a single cluster with $R_{II} = 0$. This is a third regime, distinct from both species coexistence and drift monoculture.

This matters methodologically. “No detected isolation” does not mean that the evolutionary equilibrium is uniform or unstructured. A population can be internally niche-bucketed, with per-niche specialists coexisting without cross-niche reproductive barriers. R_{II} alone cannot distinguish niche-bucketed pseudo-uniformity from true monoculture; raw pairwise compatibility distributions help, but future work should add per-agent multi-niche fitness profiles to RCM/RCC diagnostics.

6.2 Speciation cost in the LLM substrate

Track B reveals an important side contribution: enforcing reproductive isolation imposes a population-level fitness cost in our LLM-agent substrate. The hand-seeded species-coexistence regime reaches max $R_{II} = 1.0$ but has population mean fitness 3.41. In contrast, the soft-interface population-uniform regimes reach high fitness across all three niches, with means 4.36 and 4.50 in the two low-mutation settings. This is a 22–32% aggregate fitness advantage over the species-coexistence positive control.

The per-agent follow-up sharpens this claim. Across 480 agent-checkpoints, no individual agent is top-3 or top-5 across all niches. The population-uniform state is therefore not a single-agent superintelligence result; it is a specialist-per-bucket collective equilibrium. LLM-agent populations can remain one reproductive compatibility class while distributing competence across niche buckets.

This contrasts with the usual intuition from biological speciation, where reproductive isolation is often associated with enhanced local adaptation. In our LLM substrate, hard-interface constraints trade away population-level performance for measurable species boundaries. Soft interfaces allow the population to remain one recombining compatibility class while becoming broadly competent across niches. Put differently, LLM-agent populations may prefer to be one species of collectively generalist agents rather than multiple reproductively isolated specialist lineages, and forcing them apart costs aggregate performance.

The conclusion is necessarily population-level. Future RCM variants should log per-agent multi-niche profiles throughout evolution to separate individual generalists from niche-bucketed collectives in a single unified diagnostic.

6.3 Why LLM substrates need hard interfaces

Biological systems inherit hard biochemical interfaces: proteins fold into shapes, regulatory interactions depend on chemistry, and developmental constraints can make cross-lineage hybrids inviable. LLM agents inherit fluid prompt interfaces: a shared foundation model can reinterpret, translate, or repair many mismatched instructions at inference time. This interface fluidity prevents Darwinian post-zygotic isolation from appearing automatically. A prompt mismatch is often not a lethal incompatibility; it is an instruction the model can still parse.

Hard interface constraints convert this fluid substrate into one where reproductive isolation becomes measurable. Rigid mode operationalizes post-zygotic incompatibility by mapping strict-edge mismatch to zero viability. Under that condition, RCM/RCC detects species-like boundaries. Under soft penalties, between-lineage hybrids retain enough viability to keep the compatibility graph connected.

6.4 Spectral bistability under fine-grained measurement

The fine-grained RCC result shows that even hard-interface isolation can be intermittently detected under evolutionary pressure. Coarse checkpoints initially suggested collapse near generation 20, but every-generation measurement reveals post-generation-12 bistability: HOT detections persist 43% of the time with mean HOT RII = 0.97. The pairwise compatibility pattern is consistent with a persistent boundary signal because K_b remains approximately zero even in cold states; the detector flickers because K_w crosses the post-hoc validity threshold.

This reframes Empirical Observation E1. E1 is not an intrinsic collapse-timescale hypothesis. It is an empirical observation about spectral bistability under finite- N and noisy hybrid evaluation. Theorem 1 predicts recombination load from lineage-divergent edges, but it does not model finite- N spectral detectability or HOT/cold alternation. A falsifiable next step is to measure RCC at finer resolution or with repeated RCM samples at fixed checkpoints; the bistability hypothesis predicts a stable alternation rate set by hybrid-evaluation variance and within-lineage signal strength.

6.5 Measurement-conditional reproductive isolation

A key conceptual contribution is that reproductive isolation is measurement-conditional in LLM-agent systems. In biology, reproductive isolation is observed through infertility or hybrid inviability. In LLM agents, the analogue depends on the rigidity of the type-checking regime used during hybrid evaluation. With a soft penalty, mismatch reduces quality but does not eliminate hybrid viability; the resulting $K_w - K_b$ gap is too small for RCC to separate lineages. With rigid mode, a strict type-edge mismatch sets viability to zero, creating a sharp reproductive boundary. This is not a bug in the protocol. It defines what kind of incompatibility is being operationalized and therefore what kind of agent “species” can be measured.

For AI agent ecosystems, the implication is practical. If designers want independent agent lineages to remain recombinable, the natural flexibility of LLM substrates may help. If they want stable separated populations, they need explicit interface or protocol incompatibilities, plus mechanisms that actively preserve those boundaries over evolutionary time. Niche rewards and diversity objectives alone are not enough.

7 Reproducibility Statement

We will release the SAET code, founder genomes, task samplers, run manifests, and plotting scripts upon acceptance. The current experiments use a single LLM family, Azure gpt-5.4-mini, with temperature 0.7 and maximum generation length 4096 tokens. The primary founder is the typed v4 modular genome with $L = 12$ modules; its JSON schema and the three hand-seeded strict-interface lineages are included in the supplementary material. The three evaluation niches sample PlanBench Blocksworld, LoCoMo, and HotpotQA tasks, with 10 sampled tasks per niche per generation in the main runs.

RCC uses spectral eigengap clustering over the reproductive compatibility matrix with $R = 3$ hybrid offspring per ordered parent pair, $\tau_{in} = 0.10$, and $\tau_{out} = 0.05$ as post-hoc validity filters. RCM evaluation crosses agents at measurement time regardless of whether the population evolution used crossover. We evaluate different parameter settings of the same SAET pipeline rather than treating each setting as an independent method. The cumulative LLM compute cost for the experiments reported here is under \$400, well within the project budget of \$2,460.

8 Limitations

This is an exploratory single-seed study. We deliberately do not claim multi-seed statistical generalization for the LLM-agent runs. The synthetic experiments test the mathematical mechanism under controlled DMI assumptions, but those assumptions are only partially realized in the LLM substrate. Future work should repeat the full run suite across multiple seeds, LLM families, and founder designs.

The hard-interface result is a positive control, not an emergent-speciation discovery. The lineages are hand-seeded with hard strict-edge incompatibilities, and rigid mode mechanically sets cross-lineage

mismatch viability to zero. This validates that RCM/RII/RCC can detect reproductive isolation when it exists, but it does not show that such isolation emerges spontaneously or is maintained without active constraints.

The reproductive-isolation measurements are conditional on the measurement protocol. Soft mismatch penalties leave enough cross-lineage viability to blur compatibility boundaries, while rigid mismatch penalties create separable clusters. We include a post-hoc RCC threshold sensitivity sweep and raw-pairwise compatibility plots, but these are re-analyses of existing RCM matrices rather than new replicated runs.

RCM uses $R = 3$ hybrid offspring per ordered parent pair for cost feasibility. This may be insufficient for stable spectral inference; the HOT/cold alternation observed in fine-grained measurement may include sampling-variance contributions. Future work should repeat RCM at fixed checkpoints with larger R to separate biological-style compatibility structure from estimator noise.

The empirical substrate is narrow. We use one cost-optimized LLM family, Azure gpt-5.4-mini, and three no-docker niches: PlanBench Blocksworld, LoCoMo, and HotpotQA. We do not test web or code benchmarks, larger frontier models, or niche-breadth scaling. The founder genome has $L = 12$ modules in the typed v4 design; we do not perform an LLM-side module-count sweep, so the synthetic L_c scaling should be interpreted as mechanistic support rather than a direct LLM scaling law.

Empirical Observation E1 is post hoc. Our fine-grained RCC run corrects the initial coarse-checkpoint interpretation: five-generation RCC can make bistable HOT/cold dynamics look like terminal loss of structure. Other parameter settings of the SAET pipeline are supported by retroactive coarse-pattern analysis and would benefit from the same fine-grained measurement.

The spectral-bistability interpretation is supported by our fine-grained RCC run, but the architectural mechanism behind HOT/cold alternation is not directly measured. We did not track type-label entropy, strict-edge survival, suffix recoverability, or repeated RCM variance at each checkpoint. A full validation would require fine-grained RCC for every condition and repeated RCM sampling per checkpoint, which was outside the current budget.

Finally, the biological analogy is operational. We borrow the Darwinian and biological-species language to define variation, heredity, selection, and post-zygotic incompatibility in an engineered system. LLM agents do not naturally mate, and their interfaces can be made soft or rigid by experimental design. The paper’s claims concern this operational reproductive-compatibility assay, not natural biological reproduction.

We tested individual super-agent presence with a Tier A per-agent multi-niche evaluation over two low-perturbation configurations, 10 checkpoints, and 24 agents per checkpoint (480 agent-checkpoints). No single agent dominates all niches under top-3 or top-5 criteria. The super-agent regime in this paper is therefore a population-level property with specialist-per-bucket internal structure, not direct evidence for one universally dominant individual. Future work should log per-agent multi-niche fitness profiles throughout the main evolutionary runs rather than only in follow-up evaluation.

9 Conclusion

We find that LLM-agent reproductive isolation is possible but constrained. Synthetic experiments validate EST, seven spontaneous LLM forks fail to produce isolation, hand-seeded rigid incompatibilities provide a positive control with max RII = 1.0, and mild evolution makes that isolation intermittently detectable under fine-grained RCC. The resulting picture is not simply negative: it characterizes the operational regime in which LLM-agent species can exist. Soft interfaces preserve compatibility; rigid interfaces can create species-like boundaries; sustained isolation requires mechanisms that actively maintain those boundaries and measurement protocols fine-grained enough to distinguish bistability from collapse.

References

Christopher Berner, Greg Brockman, Brooke Chan, et al. Dota 2 with large scale deep reinforcement learning. *arXiv preprint arXiv:1912.06680*, 2019.

- Alastair Channon. Improving and still passing the alife test: Component-normalised activity statistics classify evolution in geb as unbounded. *Artificial Life*, 9(2):173–187, 2003.
- Jerry A. Coyne and H. Allen Orr. *Speciation*. Sinauer Associates, Sunderland, MA, 2004.
- Charles Darwin. *On the Origin of Species by Means of Natural Selection, or the Preservation of Favoured Races in the Struggle for Life*. John Murray, London, 1859.
- Stephane Doncieux, Nicolas Bredeche, Jean-Baptiste Mouret, and A. E. Eiben. Evolutionary robotics: What, why, and where to. *Frontiers in Robotics and AI*, 2:4, 2015.
- Chrisantha Fernando, Dylan Banarse, Henryk Michalewski, et al. Promptbreeder: Self-referential self-improvement via prompt evolution. *arXiv preprint arXiv:2309.16797*, 2023.
- David E. Goldberg and Jon Richardson. Genetic algorithms with sharing for multimodal function optimization. *Proceedings of the Second International Conference on Genetic Algorithms*, pages 41–49, 1987.
- Qingyan Guo, Rui Wang, Junliang Guo, et al. Connecting large language models with evolutionary algorithms yields powerful prompt optimizers. *arXiv preprint arXiv:2309.08532*, 2023.
- Shengran Hu, Cong Lu, and Jeff Clune. Automated design of agentic systems. *arXiv preprint arXiv:2408.08435*, 2024.
- Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, et al. Dspy: Compiling declarative language model calls into self-improving pipelines. *arXiv preprint arXiv:2310.03714*, 2024.
- Joel Lehman and Kenneth O. Stanley. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary Computation*, 19(2):189–223, 2011.
- Hod Lipson and Jordan B. Pollack. Automatic design and manufacture of robotic lifeforms. *Nature*, 406:974–978, 2000.
- Samir W. Mahfoud. *Niching Methods for Genetic Algorithms*. PhD thesis, University of Illinois at Urbana-Champaign, 1995.
- Ernst Mayr. *Systematics and the Origin of Species from the Viewpoint of a Zoologist*. Columbia University Press, New York, 1942.
- Charles Ofria and Claus O. Wilke. Avida: A software platform for research in computational evolutionary biology. *Artificial Life*, 10(2):191–229, 2004.
- Justin K. Pugh, Lisa B. Soros, and Kenneth O. Stanley. Quality diversity: A new frontier for evolutionary computation. *Frontiers in Robotics and AI*, 3:40, 2016.
- Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. *arXiv preprint arXiv:2303.11366*, 2023.
- Oriol Vinyals, Igor Babuschkin, Wojciech M. Czarnecki, et al. Grandmaster level in starcraft ii using multi-agent reinforcement learning. *Nature*, 575:350–354, 2019.
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, et al. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291*, 2023.
- Rui Wang, Joel Lehman, Jeff Clune, and Kenneth O. Stanley. Paired open-ended trailblazer (poet): Endlessly generating increasingly complex and diverse learning environments and their solutions. In *Proceedings of the Genetic and Evolutionary Computation Conference*, pages 142–151, 2019.
- Mert Yuksekgonul, Federico Bianchi, Joseph Boen, et al. Textgrad: Automatic differentiation via text. *arXiv preprint arXiv:2406.07496*, 2024.

A Proofs for Epistatic Speciation Theory

This appendix gives the proofs omitted from Section 3. We use the same notation as the main text. Unless otherwise stated, $F = F_\rho$ for a fixed ecological mixture, all expectations include crossover and task stochasticity, and $F(G) \in [F_{\min}, F_{\max}]$.

A.1 Auxiliary load identity

Lemma 1 (Single-edge expectation). *Consider an epistatic edge (ℓ, r) between parental alleles (a_ℓ, a_r) and (b_ℓ, b_r) . Suppose each endpoint is assigned independently as $Z_\ell \in \{A, B\}$ with $\Pr(Z_\ell = A) = \Pr(Z_\ell = B) = 1/2$ (and likewise for Z_r). Thus $\Pr[(Z_\ell, Z_r) = (A, A)] = \Pr[(Z_\ell, Z_r) = (B, B)] = 1/4$, making the same-lineage assignments symmetric, and conditional on $Z_\ell \neq Z_r$, the mixed orientations (A, B) and (B, A) occur with probability $1/2$ each. Then*

$$\mathbb{E}J_{\ell r}(g_\ell, g_r) = \frac{J_{\ell r}(a_\ell, a_r) + J_{\ell r}(b_\ell, b_r)}{2} - p_{\ell r}^{\text{mix}}\delta_{\ell r}. \quad (25)$$

Proof. Let $p = p_{\ell r}^{\text{mix}} = \Pr(Z_\ell \neq Z_r)$. Conditional on no mixing, $Z_\ell = Z_r$. By marginal balance and orientation symmetry, the two same-lineage orientations (A, A) and (B, B) have equal conditional probability, so the expected epistatic term is $\{J_{\ell r}(a_\ell, a_r) + J_{\ell r}(b_\ell, b_r)\}/2$. Conditional on mixing, the two orientations (A, B) and (B, A) have probability $1/2$ each, so the expected term is $\{J_{\ell r}(a_\ell, b_r) + J_{\ell r}(b_\ell, a_r)\}/2$. Let

$$W = \frac{J_{\ell r}(a_\ell, a_r) + J_{\ell r}(b_\ell, b_r)}{2}, \quad M = \frac{J_{\ell r}(a_\ell, b_r) + J_{\ell r}(b_\ell, a_r)}{2}. \quad (26)$$

By definition, $\delta_{\ell r} = W - M$. Therefore

$$\mathbb{E}J_{\ell r}(g_\ell, g_r) = (1 - p)W + pM = W - p\delta_{\ell r} = W - p_{\ell r}^{\text{mix}}\delta_{\ell r}, \quad (27)$$

which is the desired identity. $\square$

Lemma 2 (Expected hybrid fitness identity). *Assume the single-edge conditions of Lemma 1 for all epistatic edges and assume additive utility and cost average across parents in expectation:*

$$\mathbb{E} \left[\sum_\ell u_\ell(g_\ell) - \lambda C(G^{A \times B}) \right] = \frac{1}{2} \left\{ \sum_\ell u_\ell(a_\ell) - \lambda C(G^A) + \sum_\ell u_\ell(b_\ell) - \lambda C(G^B) \right\}. \quad (28)$$

Then

$$\mathbb{E}F(G^{A \times B}) = \frac{F(G^A) + F(G^B)}{2} - \sum_{(\ell, r) \in E^{\text{epi}}} p_{\ell r}^{\text{mix}}\delta_{\ell r}. \quad (29)$$

Proof. Expand the modular landscape for the hybrid:

$$\mathbb{E}F(G^{A \times B}) = \mathbb{E} \left[\sum_\ell u_\ell(g_\ell) - \lambda C(G^{A \times B}) \right] + \sum_{(\ell, r) \in E^{\text{epi}}} \mathbb{E}J_{\ell r}(g_\ell, g_r). \quad (30)$$

By the additive averaging assumption, the first expectation equals the parental midpoint for additive and cost terms. By Lemma 1, each epistatic expectation equals the parental midpoint epistatic contribution minus $p_{\ell r}^{\text{mix}}\delta_{\ell r}$. Summing those identities over all epistatic edges yields

$$\mathbb{E}F(G^{A \times B}) = \frac{1}{2} \left\{ \sum_\ell u_\ell(a_\ell) - \lambda C(G^A) + \sum_\ell u_\ell(b_\ell) - \lambda C(G^B) \right\} \quad (31)$$

$$+ \sum_{(\ell, r)} \frac{J_{\ell r}(a_\ell, a_r) + J_{\ell r}(b_\ell, b_r)}{2} - \sum_{(\ell, r)} p_{\ell r}^{\text{mix}}\delta_{\ell r} \quad (32)$$

$$= \frac{F(G^A) + F(G^B)}{2} - \sum_{(\ell, r) \in E^{\text{epi}}} p_{\ell r}^{\text{mix}}\delta_{\ell r}. \quad (33)$$

This proves the load identity. $\square$

A.2 Proof of Theorem 1

Proof. By Lemma 2,

$$\mathbb{E}F(G^{A \times B}) = F_P - \sum_{(\ell, r) \in E^{\text{epi}}} p_{\ell r}^{\text{mix}} \delta_{\ell r}, \quad F_P := \frac{F(G^A) + F(G^B)}{2}. \quad (34)$$

Split the epistatic edge set into the lineage-divergent incompatible set $\mathcal{E}_\partial$ and its complement:

$$\sum_{(\ell, r) \in E^{\text{epi}}} p_{\ell r}^{\text{mix}} \delta_{\ell r} = \sum_{(\ell, r) \in \mathcal{E}_\partial} p_{\ell r}^{\text{mix}} \delta_{\ell r} + \sum_{(\ell, r) \in E^{\text{epi}} \setminus \mathcal{E}_\partial} p_{\ell r}^{\text{mix}} \delta_{\ell r}. \quad (35)$$

The theorem assumes $\delta_{\ell r} \geq 0$ outside $\mathcal{E}_\partial$, so the second sum is nonnegative. For every edge in $\mathcal{E}_\partial$, the theorem assumes $p_{\ell r}^{\text{mix}} \geq p_{\min}$ and $\delta_{\ell r} \geq \delta_{\min}$; hence

$$\sum_{(\ell, r) \in \mathcal{E}_\partial} p_{\ell r}^{\text{mix}} \delta_{\ell r} \geq \sum_{(\ell, r) \in \mathcal{E}_\partial} p_{\min} \delta_{\min} = M_{\text{pair}} p_{\min} \delta_{\min}. \quad (36)$$

Substituting this lower bound on load into the identity gives

$$\mathbb{E}F(G^{A \times B}) \leq F_P - M_{\text{pair}} p_{\min} \delta_{\min}, \quad (37)$$

which is the claimed linear accumulation bound. Normalizing by $F_{\max}$ gives

$$K_{AB} = \mathbb{E}[F(G^{A \times B})/F_{\max}] \leq \frac{F_P - M_{\text{pair}} p_{\min} \delta_{\min}}{F_{\max}}. \quad (38)$$

Finally, if $M_{\text{pair}} p_{\min} \delta_{\min} > F_P - \tau_v$, then $F_P - M_{\text{pair}} p_{\min} \delta_{\min} < \tau_v$, and therefore the upper bound on expected hybrid fitness is below the viability threshold. No independence among epistatic edges is required; only the local expectation identities and additive averaging assumption are used. $\square$

A.3 Proof of Theorem 2

Proof. For a transferred typed subgraph Q , define $\text{Loss}(Q) = \sum_{e \in \partial Q} Y_e X_e$. Since each summand is nonnegative and expectation is linear regardless of dependence among edges,

$$\mathbb{E}[\text{Loss}(Q)] = \sum_{e \in \partial Q} \mathbb{E}[Y_e X_e]. \quad (39)$$

For each boundary edge e , because X_e is an indicator,

$$\mathbb{E}[Y_e X_e] = \Pr(X_e = 1) \mathbb{E}[Y_e \mid X_e = 1] + \Pr(X_e = 0) \cdot 0. \quad (40)$$

The assumptions $\Pr(X_e = 1) \geq p_f$ and $\mathbb{E}[Y_e \mid X_e = 1] \geq c_f$ imply

$$\mathbb{E}[Y_e X_e] \geq p_f c_f. \quad (41)$$

Summing over $|\partial Q|$ boundary edges gives

$$\mathbb{E}[\text{Loss}(Q)] \geq |\partial Q| p_f c_f. \quad (42)$$

By definition $D_\partial(Q) = |\partial Q|/|Q|$, so $|\partial Q| = |Q| D_\partial(Q)$ and the equivalent form $|Q| D_\partial(Q) p_f c_f$ follows. If validation is rigid, a strict mismatch on any designated boundary edge can map the offspring to failure, lowering $V_{AB}(\tau_v)$. The soft inequality remains valid by viewing Y_e as the finite pre-threshold loss or by truncating hard failure to a bounded loss scale. $\square$

A.4 Proof of Theorem 3

Proof. Let the population state be $X = (G_1, \dots, G_N) \in \mathcal{G}^N$. The assumptions are: $\mathcal{G}$ is compact and convex; F is continuous, bounded, and strictly concave with a unique maximizer; reproduction forms convex hybrids $H = \alpha G_i + (1 - \alpha) G_j$; selection uses finite weights proportional to $\exp(\beta F)$; mutation has density $m(y \mid x)$ satisfying $m(y \mid x) \geq m_0 > 0$ for all $x, y \in \mathcal{G}$; and the induced population Markov kernel is aperiodic and irreducible. Bounded fitness implies

$$0 < \exp(\beta F_{\min}) \leq \exp(\beta F(G)) \leq \exp(\beta F_{\max}) < \infty. \quad (43)$$

Thus selection never assigns zero probability to any candidate solely because of fitness. Uniformly positive mutation implies that, after mutation, every offspring has a density component at least m_0 with respect to the reference measure on $\mathcal{G}$. Applied to all N coordinates, the population transition kernel has a Doeblin minorization: there exists $\eta > 0$ and a probability measure ν on $\mathcal{G}^N$ such that

$$P(X, A) \geq \eta \nu(A) \quad (44)$$

for every state X and measurable set $A \subseteq \mathcal{G}^N$. The compactness of $\mathcal{G}^N$, bounded positive selection weights, and positive mutation density ensure that η can be chosen uniformly. A Doeblin chain is uniformly ergodic; in particular it is positive Harris recurrent, aperiodic, and has a unique stationary distribution. The stated aperiodicity and irreducibility assumptions are therefore consistent with, and reinforced by, the positive-density mutation construction.

A persistent closed reproductively isolated stationary component would require a nonempty closed communicating class that cannot be reached from, or cannot reach, other regions under the population kernel. This is impossible under irreducibility and the Doeblin minorization, because every set of positive reference measure can be reached from every state with positive probability after mutation and selection. Hence the chain cannot support two or more closed stationary components corresponding to stable reproductively isolated lineages.

It remains to show that the smooth single-niche regime does not generate a DMI hybrid-loss mechanism. The no-negative-epistasis assumption states that for all parents and all $\alpha \in [0, 1]$,

$$F(\alpha G_i + (1 - \alpha)G_j) \geq \alpha F(G_i) + (1 - \alpha)F(G_j). \quad (45)$$

Taking expectations over the selected parents, the hybridization coefficient, mutation/task noise before measurement if present, and any measurement stochasticity gives

$$\mathbb{E}F(G^{a \times b}) \geq \mathbb{E}[\alpha F(G^a) + (1 - \alpha)F(G^b)]. \quad (46)$$

Assuming the hybridization coefficient is independent of the selected parental identities and their measured fitness, this becomes

$$\mathbb{E}F(G^{a \times b}) \geq \mathbb{E}\alpha \mathbb{E}F(G^a) + (1 - \mathbb{E}\alpha)\mathbb{E}F(G^b). \quad (47)$$

For symmetric hybridization, $\mathbb{E}\alpha = 1/2$, and therefore

$$\mathbb{E}F(G^{a \times b}) \geq \frac{1}{2}\{\mathbb{E}F(G^a) + \mathbb{E}F(G^b)\}. \quad (48)$$

Thus cross-lineage mixing is not systematically penalized below the parental midpoint in this regime. Combining this midpoint property with unique ergodicity and positive mutation flow proves that stable RI requires violation of at least one control assumption, such as multiple ecological niches, negative epistasis, rigid typed interfaces, restricted migration, label preservation, protocol drift, or non-smooth constraints. $\square$

A.5 Proof of Theorem 4

Proof. Let $N_L = \binom{L}{2}$ and $S_L = \sum_{i < j} X_{ij}$ with independent $X_{ij} \in [0, B]$ and $\mathbb{E}X_{ij} = \alpha p \bar{\delta}$. This specification corresponds to the DMI snowball count

$$M_{\text{pair}}(L) = \alpha \frac{L(L-1)}{2} + O(L), \quad (49)$$

with the $O(L)$ term suppressed in the leading concentration statement; retaining it changes lower-order constants but not the square-root threshold scaling. By linearity of expectation,

$$\mathbb{E}S_L = \sum_{i < j} \mathbb{E}X_{ij} = N_L \alpha p \bar{\delta} = \alpha p \bar{\delta} \frac{L(L-1)}{2}. \quad (50)$$

Because the variables are independent and each lies in an interval of range B , Hoeffding's standard bounded-range inequality gives

$$\Pr(|S_L - \mathbb{E}S_L| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i < j} B^2}\right) = 2 \exp\left(-\frac{2t^2}{N_L B^2}\right). \quad (51)$$

Equivalently, for the empirical mean $\bar{X} = S_L/N_L$,

$$\Pr(|\bar{X} - \mathbb{E}\bar{X}| \geq \epsilon) \leq 2 \exp\left(-\frac{2N_L\epsilon^2}{B^2}\right), \quad (52)$$

which is the standard form $\Pr(|\bar{X} - \mu| \geq \epsilon) \leq 2 \exp(-2n\epsilon^2/\text{range}^2)$ with no additional $1/4$ factor.

Under negligible within-lineage load, the random hybrid fitness is $F_H(L) = F_W - S_L$. Therefore the mean hybrid fitness is below the viability threshold exactly when

$$\mathbb{E}F_H(L) = F_W - \mathbb{E}S_L < \tau_v, \quad (53)$$

or equivalently

$$\alpha p \bar{\delta} \frac{L(L-1)}{2} > F_W - \tau_v. \quad (54)$$

Solving the corresponding equality for L gives

$$L^2 - L - \frac{2(F_W - \tau_v)}{\alpha p \bar{\delta}} = 0, \quad (55)$$

whose positive root is

$$\bar{L}_c = \frac{1}{2} + \sqrt{\frac{1}{4} + \frac{2(F_W - \tau_v)}{\alpha p \bar{\delta}}}. \quad (56)$$

The smallest integer complexity at or above this root is

$$L_c = \left\lceil \frac{1}{2} + \sqrt{\frac{1}{4} + \frac{2(F_W - \tau_v)}{\alpha p \bar{\delta}}} \right\rceil. \quad (57)$$

For large $(F_W - \tau_v)/(\alpha p \bar{\delta})$, the leading term is the square root, hence

$$L_c = \Theta\left(\sqrt{\frac{F_W - \tau_v}{\alpha p \bar{\delta}}}\right). \quad (58)$$

If $\Delta_L := \mathbb{E}S_L - (F_W - \tau_v) > 0$, then

$$V_{AB}(\tau_v) = \Pr(F_W - S_L \geq \tau_v) = \Pr(S_L \leq F_W - \tau_v) = \Pr(S_L - \mathbb{E}S_L \leq -\Delta_L). \quad (59)$$

The lower-tail half of Hoeffding gives

$$V_{AB}(\tau_v) \leq \exp\left(-\frac{2\Delta_L^2}{N_L B^2}\right). \quad (60)$$

This proves the concentration and the quadratic DMI complexity threshold. $\square$

A.6 Proof of Corollary 1

Proof. In the sparse bounded-degree regime, the epistatic graph has average degree $d = O(1)$, so the number of epistatic edges is $n_E = |E^{\text{epi}}| = Ld/2$. Let $X_e \in [0, B]$ be the exposed load on edge e , incorporating the probability that the edge is lineage-divergent, the crossover exposure, and the incompatibility cost. If a fraction q of realized edges can diverge and the expected exposed cost on such edges is $r\bar{\delta}$, then $\mathbb{E}X_e = qr\bar{\delta}$ under the homogeneous version of the model. Therefore

$$\mathbb{E}\mathcal{L}_{AB} = \sum_{e \in E^{\text{epi}}} \mathbb{E}X_e = \frac{Ld}{2} qr\bar{\delta}. \quad (61)$$

If the X_e are independent and bounded in $[0, B]$, Hoeffding gives

$$\Pr(|\mathcal{L}_{AB} - \mathbb{E}\mathcal{L}_{AB}| \geq t) \leq 2 \exp\left(-\frac{2t^2}{(Ld/2)B^2}\right). \quad (62)$$

With random hybrid fitness $F_H = F_W - \mathcal{L}_{AB}$, the mean threshold condition is

$$\frac{Ld}{2} qr\bar{\delta} > F_W - \tau_v. \quad (63)$$

Solving for L gives

$$L_c^{\text{sparse}} = \frac{2(F_W - \tau_v)}{dqr\bar{\delta}}. \quad (64)$$

As $d \rightarrow O(1)$, the number of interaction opportunities is $O(L)$ rather than $O(L^2)$, so the boundary case is linear. This is exactly the sparse limit of the dense snowball theory: replacing $\binom{L}{2}$ by $Ld/2$ changes the critical scaling from square-root to linear. $\square$

A.7 Proof of Theorem 5

Proof. A protocol schema $\Pi = (\mathcal{M}, \mathcal{T}, \tau_{\text{msg}})$ defines typed interfaces between emitter modules and consumer modules. In a hybrid that inherits an emitter from lineage A and a consumer from lineage B , a message boundary edge is mismatched when the emitted message type $\tau_{\text{msg}}^A(m)$ is not interpretable under the consumer’s expected type map τ_{msg}^B . Thus each emitter–consumer message channel is an edge in the typed interface boundary ∂Q of Theorem 2, with $X_e = 1$ denoting a protocol/type mismatch and Y_e denoting the downstream loss in parsing, routing, retrieval, tool invocation, or verification. Therefore, once protocol drift creates a positive mismatch probability and positive conditional loss, Theorem 2 gives

$$\mathbb{E}[\text{Loss}_{\text{protocol}}(Q)] \geq |\partial Q| p_f c_f. \quad (65)$$

This establishes the reduction of protocol incompatibility to interface-boundary hybrid loss.

It remains to justify the drift-synchronization condition. Let $D_t \in [0, D_{\text{max}}]$ be bounded protocol distance, and suppose

$$\mathbb{E}[D_{t+1} - D_t \mid D_t] \geq \sigma - cm =: \gamma > 0 \quad (66)$$

whenever D_t is below its saturation region. Assume one-step increments are bounded by b in absolute value. Then the process $D_t - \gamma t$ is a submartingale up to saturation. Azuma–Hoeffding implies that for T pre-saturation steps,

$$\Pr(D_T - D_0 \leq \gamma T - s) \leq \exp\left(-\frac{s^2}{2Tb^2}\right). \quad (67)$$

Choosing $s = \gamma T/2$ shows that, with probability at least $1 - \exp(-\gamma^2 T/(8b^2))$, protocol distance increases by at least $\gamma T/2$ before saturation. If cross-protocol accuracy is Lipschitz decreasing in distance, for example

$$A_C(i, j) \leq A_C(i, i) - \kappa D_T \quad (68)$$

for some $\kappa > 0$ on the relevant range, then increasing D_T lowers $A_C(i, j)$ and $A_C(j, i)$ relative to within-lineage accuracies. Consequently

$$I_C(i, j) = 1 - \frac{A_C(i, j) + A_C(j, i)}{A_C(i, i) + A_C(j, j) + \varepsilon} \quad (69)$$

increases with high probability over sustained divergence. Since these protocol mismatches are typed boundary mismatches, the resulting loss reduces to Theorem 2; hence protocol isolation is an interface-boundary special case. $\square$

B Run Registry

The spontaneous-failure suite contains seven verified SAET variants. We tested interventions across genome complexity, assortativity, mutation intensity, migration, reward shaping, interface rigidity, and mixed strategy combinations. All seven spontaneous variants returned no detected reproductive-isolation structure: the final compatibility partition was $K = 1$ with $\text{RII} = 0$. In addition to these spontaneous variants, the paper reports a hand-seeded hard-interface positive control, several mild-evolution configurations of the same SAET pipeline, soft-mode operational baselines, and a single-niche anchor. We omit per-run seed details from the main text because the study is exploratory and single-seed; full manifests will be released with the code upon acceptance.

C Spontaneous Failure Runs

The spontaneous-failure suite tests seven SAET variants spanning genome complexity, assortativity, mutation intensity, migration, reward shaping, interface rigidity, and mixed strategies. All seven verified spontaneous variants ended with no detected reproductive-isolation structure: final $K = 1$ and $\text{RII} = 0$. These results are supporting evidence for the main contrast in Section 5.2: hard interfaces create detectable boundaries, whereas ordinary ecological pressure does not reliably create them.

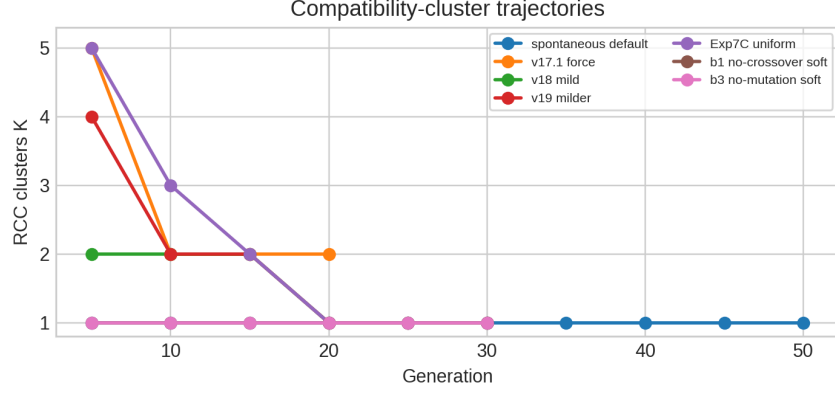


Figure 5: Compatibility-cluster trajectories for representative spontaneous and control configurations.

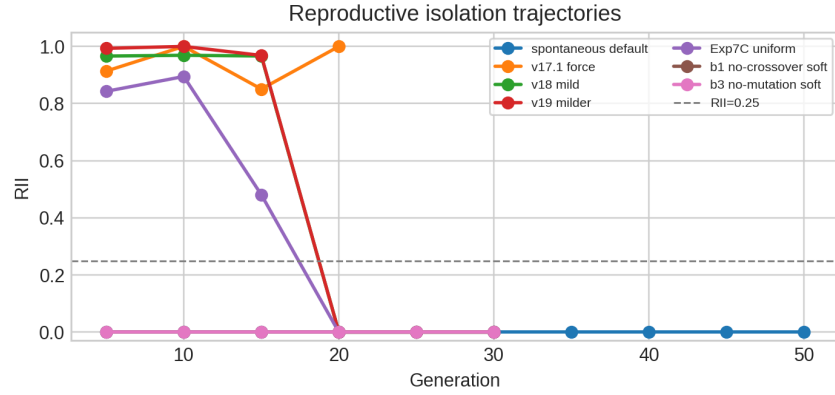


Figure 6: RII trajectories for representative configurations.

D Soft-Mode Measurement-Conditional Baselines

Soft-mode baselines demonstrate that RII is conditional on the operational rigidity of hybrid evaluation. When mismatch only applies a soft penalty, cross-lineage hybrids retain enough viability to blur spectral boundaries, even when population crossover or mutation is disabled.

E Bistability and Mild-Evolution Ablations

Coarse five-generation sampling can make spectrally bistable dynamics look like terminal loss of structure. The fine-grained run in Section 5.4 motivates reinterpreting first-cold checkpoints as undersampled HOT/cold alternation.

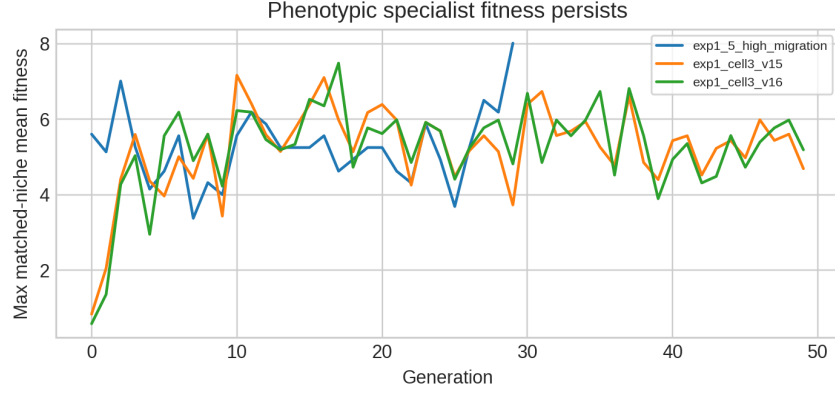


Figure 7: Phenotypic specialist fitness can persist even without detected reproductive isolation.

Table 5: Soft-mode operational baselines.

Exp.	Variant	Crossover	Setting	Outcome
soft A	per-niche	disabled	mild evolution	$K = 1$; $K_w = 0.08\text{--}0.15$; $\text{RII} = 0$
soft B	fixed template	typed	no mutation	$K = 1$; $K_w = 0.10\text{--}0.16$; $\text{RII} = 0$

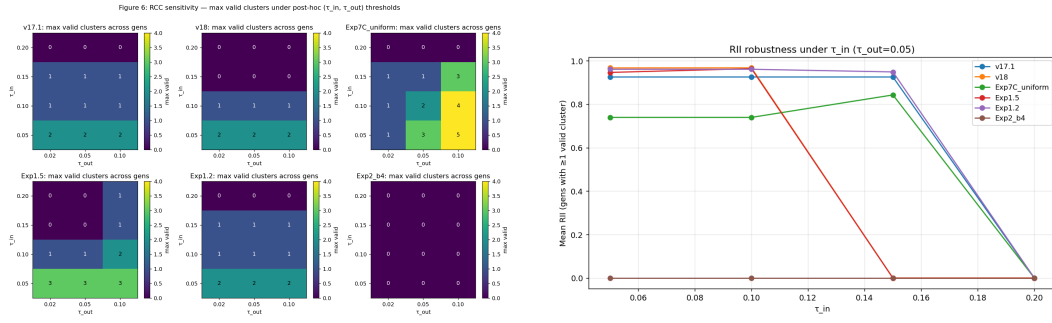


Figure 8: RCC sensitivity to post-hoc validity filters. The qualitative conclusions are stable across tested τ_{in} and τ_{out} values.

Table 6: Coarse five-generation observations before fine-grained measurement. The first-cold checkpoint should be interpreted as undersampled bistability.

Run	Gen 5 RII	Gen 10 RII	Gen 15 RII	First cold
mild config A	0.966	0.969	0.967	20
mild config B	0.993	1.000	0.969	20
uniform-crossover config	0.840	0.895	0.478	20
mild config C	1.000	0.885	0.970	30

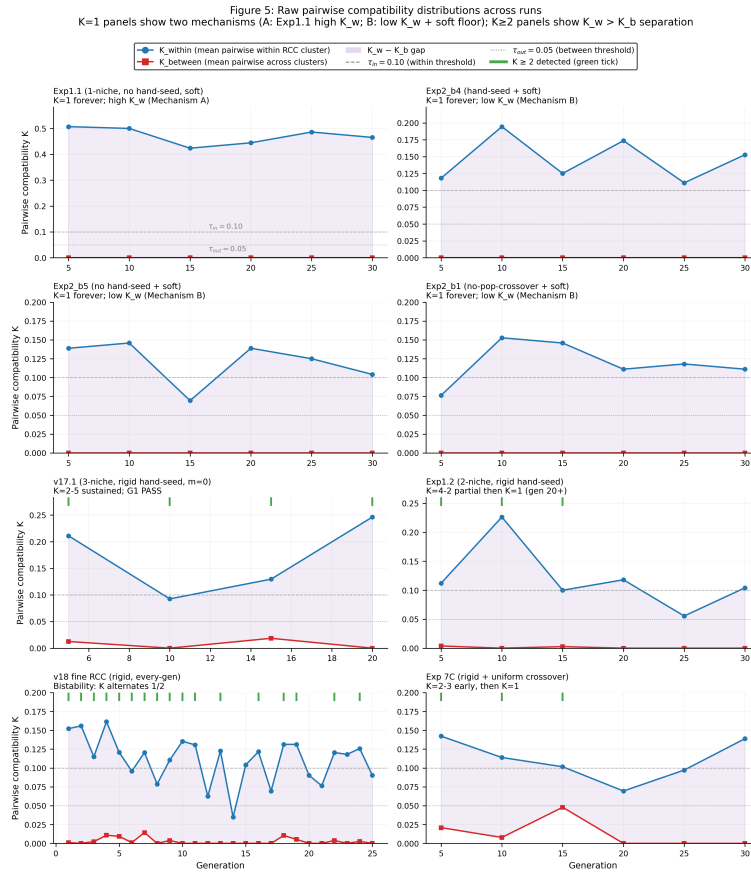


Figure 9: Faceted raw pairwise compatibility distributions. $K = 1$, $R_{II} = 0$ can indicate high compatibility or low sparse signal, depending on the run.

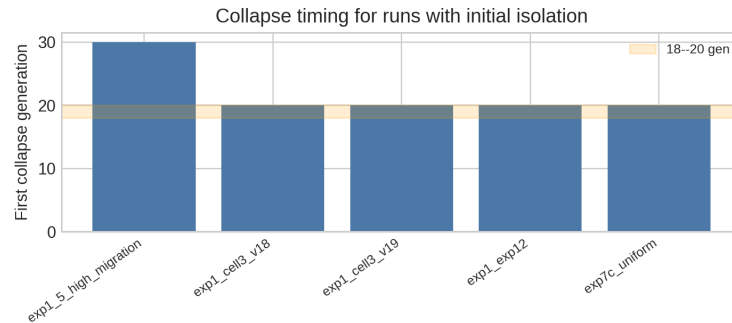


Figure 10: First-cold checkpoints under coarse five-generation sampling. Fine-grained measurement shows that these checkpoints can reflect spectral bistability rather than terminal loss of structure.